

AdaBoost **combines weak learners into a stronger classifier** by adding weighted votes. For $y_i, h_t(x) \in \{-1, +1\}$, start with $F_0 = 0$ and update

$$F_t(x) = F_{t-1}(x) + \alpha_t h_t(x), \quad \hat{y}(x) = \text{sign}(F_T(x)).$$

At the start of round t , define each observation's potential, the **total potential** Z_t , and the normalized training weights: $w_t(i) = e^{-y_i F_{t-1}(x_i)}$, $Z_t = \sum_{i=1}^n w_t(i)$ and $D_t(i) = w_t(i)/Z_t$. So $\{D_t(1), D_t(2), \dots, D_t(n)\}$ forms a probability vector.

By definition, we have $Z_1 = n$ and $D_1(i) = 1/n$. Poorly classified observations have larger potentials and receive more weight. Round t changes the total potential from Z_t to Z_{t+1} .

Choose a weak learner h_t such that the weighted error

$$\epsilon_t = \sum_i D_t(i) \mathbf{1}\{h_t(x_i) \neq y_i\}.$$

We assume $0 < \epsilon_t < \frac{1}{2} - \delta_t$ for some $\delta_t > 0$. So δ_t denote the learner's advantage over a random guess.

For a proposed coefficient α , the update gives $w_{t+1}(i) = w_t(i)e^{-\alpha y_i h_t(x_i)}$. Separating correct and incorrect predictions yields

$$\frac{Z_{t+1}}{Z_t} = \sum_i D_t(i) e^{-\alpha y_i h_t(x_i)} = (1 - \epsilon_t) e^{-\alpha} + \epsilon_t e^{\alpha}.$$

For fixed $\alpha > 0$, this ratio increases with ϵ_t , motivating a learner with small weighted error. For fixed h_t , minimize the ratio over α :

$$-(1 - \epsilon_t)e^{-\alpha} + \epsilon_t e^{\alpha} = 0 \implies \alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t}.$$

Substitution gives the **per-round potential bound**

$$\frac{Z_{t+1}}{Z_t} = 2\sqrt{\epsilon_t(1 - \epsilon_t)} = \sqrt{1 - 4\delta_t^2} \leq e^{-2\delta_t^2},$$

where the inequality follows from $\log(1 - u) \leq -u$. The normalized weights for the next round are

$$D_{t+1}(i) = D_t(i) e^{-\alpha_t y_i h_t(x_i)} \frac{Z_t}{Z_{t+1}}.$$

Incorrect predictions gain weight relative to correct predictions. Here Z_t is the total potential; the update's normalization factor is the *ratio* Z_{t+1}/Z_t .

After T rounds, the final potential is Z_{T+1} . Using $\mathbf{1}\{y_i F_T(x_i) \leq 0\} \leq e^{-y_i F_T(x_i)}$ and $Z_1 = n$,

$$\hat{R}_{\text{train}}(F_T) \leq \frac{Z_{T+1}}{n} = \prod_{t=1}^T \frac{Z_{t+1}}{Z_t} \leq \exp\left(-2 \sum_{t=1}^T \delta_t^2\right).$$

If every round has $\delta_t \geq \delta > 0$, this is at most $e^{-2T\delta^2}$. Repeated weak advantages therefore yield a decreasing training-error bound.